SHLOK JAIN

Senior Undergraduate
Department of Computer Science Engineering
Indian Institute of Technology Kanpur

Shlok-Jain
Shlok Jain
+91-9316727704
jainshlok23@iitk.ac.in

Year	Degree/Certificate	Institute	CPI/%
2023 – Present	B.Tech	Indian Institute of Technology Kanpur	**9.7**/10
2023	GSEB(XII)	Divine Life School, Ahmedabad	85.69%
2021	GSEB(X)	Divine Life School, Ahmedabad	91.16%

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 177** in JEE Advanced 2023, conducted by IIT Guwahati amongst the 250,000 shortlisted candidates.
- Secured **All India Rank 268** amongst the approximately 1 million applicants in JEE Mains 2023, conducted by the NTA.
- Awarded the **Academic Excellence Award** for exceptional academic performance for **three years**
- Actively participate in competitive programming and achieved the title of **Expert** on **Codeforces**
- Qualified **Kishore Vaigyanik Protsahan Yojna (KVPY)** 2021, conducted by IISc Banglore with **AIR 863**

WORK EXPERIENCE

Quadeye Securities LLP — *May'26 – Jul'26*
Systems Engineer Intern

- Developed an end-to-end **DAG orchestration** framework for defining, and executing complex data pipelines with multiple executors, **isolated pipeline runs**, and **output reuse** and **caching**
- Built a declarative **JSON-based pipeline** interface with configurable nodes, edges, and custom **handler functions**
- Implemented **PostgreSQL**-backed orchestration engine with DB-triggered dependency tracking, failure propagation, **runtime CSV generation** using **cloudpickle**
- Designed and implemented a caching module to work at **proc level** to cache a particular execution, considering **eBPF based** access tracking for consistency and correctness
- Developed the caching module in an optimised manner to **reduce latency** using efficient hashing algorithms like **blake3**, to reduce wasted **compute cost** and **time**

SuperLiving — *Jan'26 – Apr'26*
AI Engineer Intern

- Worked on developing a hybrid **AI memory architecture** for various Lifestyle development chatbots to improve UX
- Explored and experimented with various naive solutions based on **cosine-similarity search**, to **complex graph-based memory architectures** like **Mem0** and **Zep** in Neo4j graph database
- Implemented an architecture based on **long-term**, **short-term**, **episodic**, as well as **calender based memories**, efficient storage and retrieval workflows, also solving problems like bolating
- Extensive **benchamarking** and testing using **open-source datasets** like **LoCoMo** and **LongMemEval**
- Analyzed the failure cases from benchmarking, further tweaking and fine-tuning the system, achieving **accuracy score of 64%** comparable to existing solutions like **Mem0** an **OpenAI Memory**

COMPETITIONS

ISRO Geospatial — Silver (2nd place) — *Nov'25 – Dec'25*
InterIIT Tech Meet 14.0, IIT Patna

- **Published research paper** on this work, accepted at **IGARSS Conference** in **Washington, D.C.**
- Built a unified multimodal pipeline supporting **SAR**, **FCC** and **NCC** imagery, achieving **98.98% modality-classification accuracy** and robust downstream image understanding
- Fine-tuned **Qwen3-VL, YOLO11-OBB, SAM2.1/SAM3** and **LLaMA-3.1**; developed Tri-YOLO + GeoQwen grounding pipeline, improving VRSBench score to **0.57**
- Engineered geometric and temporal reasoning modules using **SAM2.1/SAM3 + RemoteCLIP** and **BiSAM**, enabling object measurement, counting, area/perimeter estimation and **satellite-image change detection**

KEY PROJECTS

Verilog FPGA Processor — *Jan'25 – Apr'25*
CS220 Course Project, Prof. Debapriya Basu Roy

- Designed a **32-bit single cycle MIPS processor** using **Verilog**, implemented on **Xilinx PYNQ-Z2 FPGA** board
- Designed **instruction encoding** similar to MIPS, ensuring **IF, ID, EX, MEM, and WB** cycles work properly
- Implemented a subset of instructions including **floating point operations**; tested **Insertion Sort** on hardware

Transaction support for FlexClone — *Jan'26 – Apr'26*
CS614 Course Project, Prof. Debadatta Mishra

- Extended **FlexClone** in the **Linux kernel** with an **ioctl-driven transactional framework**, providing atomic multi-block writes, inode-level mutual exclusion, and isolation from child snapshots
- Implemented versioned reads using an **append-only log** and **XArray**-based in-memory index, with on-demand version loading and **FIFO eviction** to reduce read-time overhead
- Engineered **crash-consistent** recovery using **CRC verification**, rollback of incomplete versions, and **post-mount consistency checks**; additionally implemented **garbage collection** for reclaiming orphaned blocks

Parallel Computing — *Jan'26 – Apr'26*
CS633 Course Project, Prof. Preeti Malakar

- Engineered 3D stencil engine using **MPI non-blocking halo exchanges** & custom datatypes to optimize network latency
- Implemented **point-to-point parallel data exchange** protocols, mitigating synchronisation bottlenecks & serialisation
- Profiled distributed execution scaling up to **96 processes**, diagnosing inter-node overhead and shared resource contention

Building GemOS — *Aug'25 – Nov'25*
CS330 Course Project, Prof. Debadatta Mishra

- Built a **grep-like tool**, **syscall tracer**, and **process chaining shell** using strace, file I/O, fork/exec, and pipes
- Handled memory via **mmap/mremap** and designed custom malloc/free to track heap pages and manage fragmentation
- Implemented **round-robin scheduling** for fair CPU allocation; performed multi-level page table walks for mapping inspection
- Implemented multithreading, semaphore based synchronisation with **spinlocks**, wait queues and **copy-on-write fork**

TECHNICAL SKILLS

Languages, Libraries & Skills: C | C++ | Python | JS | Verilog | MPI | Bash | Git | ReactJS | NodeJS | NumPy | Pandas | Scikit-learn | Matplotlib | Streamlit | NLTK | LLMs | TensorFlow | OpenCV

RELEVANT COURSES

Operating Systems	Probability for CS
Computer Organisation	Computer Networks
Linux Kernel Programming	Intro to Machine Learning
Financial Econometrics	Software Development
Theory of computation	Discrete Mathematics
Parallel Computing	Data Structures & Algorithms
Logic for CS	Linear Algebra and ODE

POSITIONS OF RESPONSIBILITY

Leader, IITK Consulting Group — *Apr'25 – Mar'26*

- Spearheaded a 2-tier team of **40 students** through partnerships with **10+** social organizations, providing pro-bono services
- Worked on impactful projects with organisations like **UNICEF**, **Pratham International**, **AP Govt**, and **UP Govt**